R Code

```r
# ========================================
#           Factor Analysis
# ========================================

# 1. Load dataset
data(mtcars)

# 2. Select variables
data_fa <- mtcars[, c("mpg","disp","hp","drat","wt","qsec")]

# 3. Standardize data
data_scaled <- scale(data_fa)

# 4. Correlation matrix
cor_matrix <- cor(data_scaled)
print("Correlation Matrix:")
print(cor_matrix)

# 5. Eigenvalues
eigen_values <- eigen(cor_matrix)$values
print("Eigenvalues:")
print(eigen_values)

# 6. Scree Plot
plot(eigen_values,
     type="b",
     main="Scree Plot",
     xlab="Factor Number",
     ylab="Eigenvalue")

# 7. Perform Factor Analysis
fa_result <- factanal(data_scaled, factors=2, rotation="varimax")

print(fa_result)

# 8. Extract factor loadings
loadings_matrix <- as.matrix(fa_result$loadings)

# 9. Compute communalities
communalities <- rowSums(loadings_matrix^2)

# 10. Extract specific variance (uniqueness)
specific_variance <- fa_result$uniquenesses

# 11. Create final factor analysis table
fa_table <- cbind(loadings_matrix,
                  Communality = communalities,
                  Specific_Variance = specific_variance)

print("Final Factor Analysis Table:")
```

```r
print(round(fa_table,3))

#====================Practice Questions===================================

#Q1. Perform factor analysis for USArrests data in R.
#Q2. Perform factor analyis for "big_five_personality_teaching_dataset".
```